GOMBARAGODAGE KAVEESHA DHANANJAYA

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kd-27.github.io/portfolio · github.com/KD-27 · Google Scholar

RESEARCH INTERESTS

Model predictive control (MPC), learning-augmented control and motion planning for robots that move through, and physically interact with, unstructured environments. State estimation and perception in the loop, using vision, LiDAR and force sensing to estimate what cannot be measured directly. Applied across aerial, ground and legged platforms.

EDUCATION

BSc Engineering (Hons) in Mechatronic Engineering 2021 – 2025
General Sir John Kotelawala Defence University (KDU), Sri Lanka

- First Class Honours with a cumulative GPA of 3.91 / 4.00, ranked first in the cohort. Fourth-year GPA 4.00.
- The medium of instruction was English. The program lasted four years.
- Selected coursework included Control Systems Engineering (A+), Advanced Control Systems (A+), Non-Linear Control Systems (A+), Intelligent Systems Control (A+), Signals and Systems (A+), Digital Signal Processing (A+), Applied Statistics (A+), Advanced Calculus (A+), Microprocessors and Embedded Systems, Machine Learning, Power Electronics, Electrical Machines and Drives, Analog and Digital Electronics.

PUBLICATIONS

Under review

- M. Thilakarathna, **G. K. Dhananjaya**, A. Wijesinghe, M. Vaszilijevics, E. Hood, and D. Herath, "FusionGrasp: A Modular Framework for Integrating Heterogeneous 6D Grasp Planners for Multimodal Cluttered Sorting," *Australasian Conference on Robotics and Automation (ACRA)*. Under review.
  Second author: wrote the paper.

Published

- **G. K. Dhananjaya**, J. A. C. A. Perera, G. R. A. Senarathne, J. A. D. I. Priyashan, and B. Hettige, "Prediction of Coronary Artery Disease Using Artificial Neural Network," *KDU 17th International Research Conference*, 2024. (Oral presentation.) DOI 10.64701/irc2024/345/8770.
  First author: built the neural network and wrote the whole paper.
- M. L. Saputantri, **G. K. Dhananjaya**, J. A. C. A. Perera, and U. A. N. D. Ubeysiriwardana, "Development of an Automated Clothesline System," *KDU 16th International Research Conference*, 2023. (Poster presentation.)
  Second author: built the sensor and control setup and wrote some sections of the paper. Fabrication was done by a co-author.

RESEARCH EXPERIENCE

Quadruped Robot for Inspection in Human-Inaccessible Environments 2024 – 2025
Final Year Group Research Project | Project lead | Supervised by Mr. Isira Naotunna

- Designed five-bar parallel legs to reduce reflected inertia and simplify actuation.
- Derived closed-form inverse kinematics using the Variable Circle Method, eliminating the need for iterative solving.
- Optimised gait trajectories in MATLAB and validated them in simulation.
- Implemented distributed ROS 2 / micro-ROS control across the onboard computer and microcontroller.

LiDAR-Based SLAM for Autonomous Mapping and Navigation 2023
Third Year Research Project | Supervised by Mr. Isira Naotunna

- Designed and fabricated a rocker-bogie-inspired wheeled platform with 2D LiDAR for autonomous indoor navigation.
- Implemented a ROS SLAM and navigation pipeline that maintained position and heading errors within 10 cm and 15°, respectively.
- Modelled the platform kinematics and validated the SLAM and navigation stack in Gazebo before hardware deployment.

Lazy Coulomb Planner, a reactive path planner based on electrostatic repulsion 2026
Independent project | github.com/KD-27/Lazy-Coulomb-Planner (humble branch)

- Developed a planner that starts with a straight-line path and adds repulsive waypoints only at obstacle intersections instead of evaluating a continuous potential field.
- Implemented perpendicular escape motion and waypoint locking to reduce oscillation and local-minimum trapping.
- Built the planner as a ROS 2 Nav2 plugin and tested it on a 1.8 m-tall mobile robot. Documented a qualitative comparison with APF, RRT, A*, VFH and Bug algorithms, with limitations in narrow passages, concave obstacles and path optimality.

PROFESSIONAL EXPERIENCE

Senior Robotics Engineer Sep 2025 – Present
Hype Insight (Pvt) Ltd, Crows Nest, NSW, Australia. Based at Hype Invention, Colombo 02, Sri Lanka

Aerial autonomy stack for predictive control, planning and dynamic environments

- Integrated global mapping and planning with MPC and PX4, then validated the stack in multi-level PX4/Gazebo environments.
- Implemented moving-obstacle filtering with vehicle yielding, replanning and 3D routing over simulated crowd regions in PX4/Gazebo.
- Developed frontier-based exploration for autonomous mapping of unknown indoor spaces.

Autonomous ground platform for perception, planning and hardware

- Lead engineer for a four-wheel autonomous robot on an NVIDIA Jetson AGX Orin 64 GB, integrating ZED X, OAK-D Pro Wide and RPLIDAR S2/A2 with BLDC drives, servos and a motion-compensated DSLR and gimbal payload over a shared CAN bus.
- Implemented autonomous navigation in crowded, human-shared environments using Voronoi zoning, ground-plane rejection and removal of pedestrian artefacts from SLAM maps.
- Designed sheet-metal chassis, large-format FDM structures and composite lay-ups for improved finish and stiffness.

Embedded wireless audio system development

- Developed nRF52840 firmware in Zephyr RTOS for continuous encrypted audio capture to onboard flash and completed an 85-minute, 256,156-frame recording with no dropped or corrupted frames.
- Implemented sustained BLE transfers at 113 KB/s with resumable downloads and built a Kotlin Android client for retrieval, decryption and playback.

Product Development Engineer Apr 2025 – Aug 2025
MAS Capital (Pvt) Ltd, Intimates, Ratmalana, Sri Lanka

- Designed and tested pneumatically powered moulds to form pallets from styrene-ethylene-butylene-styrene (SEBS), a thermoplastic elastomer, evaluating 8–10 formulations for performance and recovery.
- Researched and tested polyurethane reactive (PUR) adhesives for two heated-gel deposition machines (FLOD and DOD) to assess bonding feasibility for production.
- Built prototypes using knife-cutting, punching, laser-cutting and heat-press machines.
- Developed products for external clients through bulk manufacturing processes, combining novel materials and evaluating mechanical and functional properties against client specifications.
- Ran feasibility studies on design and process viability before prototyping, and prepared material-consumption estimates, standard minute values (SMVs), and costings across product styles.

Automation Engineering Intern Oct 2023 – Apr 2024
MAS Capital (Pvt) Ltd, Intimates, Ratmalana, Sri Lanka

- Designed the mechanical and electrical systems for a "Spaghetti Iron" machine, then validated and deployed it at the Unichela (Panadura) plant, where it remains in production.
- Automated a spaghetti cutting unit by integrating a universal tension detector, and improved an automatic sewing machine using a Megatron controller with strap-presence detection.
- Developed a convolutional neural network achieving 85 % accuracy for bra-cup orientation detection as an extension of an automatic stabiliser-attach machine.
- Programmed a UR5 robot arm for pick-and-place handling of components on the stabiliser-attach line.
- Designed base-plate and guide mechanisms for pullout and buttonhole machines, and ran a research project on electroplating of 3D-printed parts.

TEACHING EXPERIENCE

Tutor, then Head of Operations Dec 2023 – Dec 2024
360 Labs, an early-stage STEM education venture, Colombo

- Designed a hands-on robotics curriculum and STEM learning kits, and ran workshops in schools introducing students aged 5–20 to mechatronics and embedded programming.
- Promoted to Head of Operations in Sep 2024, running workshops and automation projects for agriculture.

TECHNICAL SKILLS

Robotics & Autonomy ROS 2, micro-ROS, Nav2, SLAM Toolbox, PX4, MAVROS, Gazebo, RViz, Cyclone DDS
Planning & Control MPC (OSQP), trajectory optimisation, inverse kinematics, state estimation
Perception & ML PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO
Programming C++, Python, Kotlin, MATLAB
Embedded Systems Zephyr RTOS, nRF52840, BLE, CAN bus, Jetson AGX Orin, ESP32, PLC (Siemens, Delta)
Sensors & Actuation ZED X, OAK-D Pro Wide, RPLIDAR S2/A2, Livox Mid-360, IMU, BLDC drives, gimbal systems
CAD & Electronics SolidWorks, Fusion 360, ANSYS, PCB design (KiCad, EasyEDA), LTspice
Fabrication Sheet metal, FDM, composite lay-up, laser cutting, turning and milling

HONORS & AWARDS

Gold Medal for Best Academic Performance, Mechatronic Engineering, KDU 2025
Vice-Chancellor's List, KDU (Years 1, 3 and 4)
1st Place & Best Design, KDU Mechatronics Systems Design Competition 2023

VOLUNTEER & COMMUNITY ENGAGEMENT

Community Service Project for the President's Scout Award 2020

- Organised and led a New Year celebration for residents of an elderly care home as the community service component of the President's Scout Award. Cleaned and repaired the facility, arranged traditional food, and spent the day sharing music and dance with residents.

LEADERSHIP & SERVICE

Research In-Charge, Electronic Robotics and Innovation Club, KDU 2022 – 2023
Senior Prefect, Royal College, Colombo 2021

REFERENCES

Available upon request.